Work Portfolio

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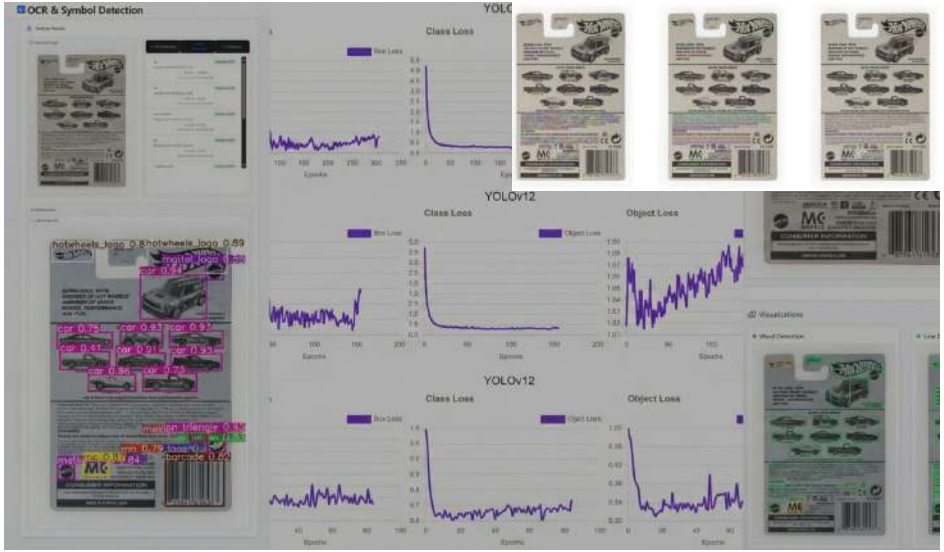
Software Engineer | AI Engineer

Product Packaging Verification System



Challenges

Manual verification is slow, error-prone, and doesn't scale

Solutions

- Automated detection of symbols & text
- AI-powered verification given requirements
- Structured compliance reports

Key Results

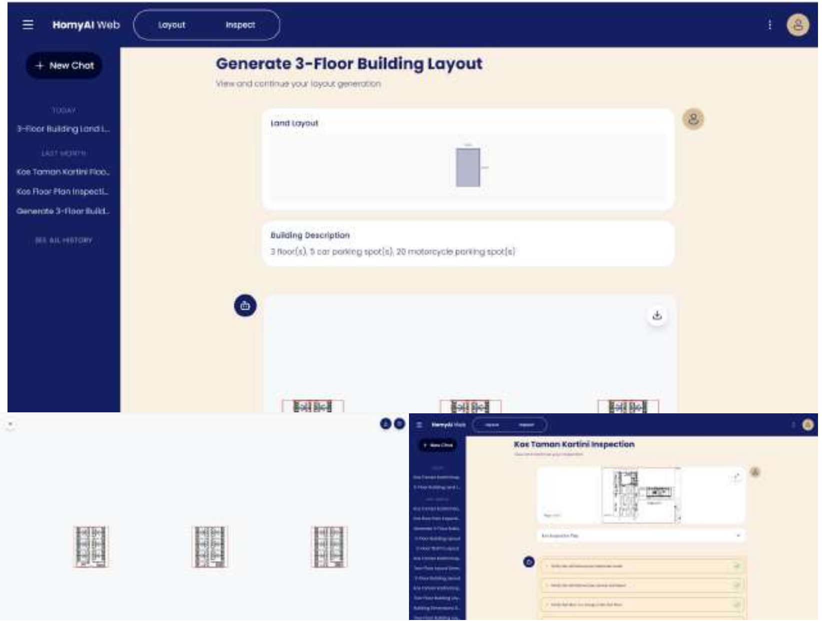
- Text Segmentation F1 Score: **85.3%**
- Symbol Detection mAP@50: **82.7%**
- Passed all given scenarios during end-to-end test

Technology: Pytorch, Roboflow, OCR, Object Detection, Scene Text Segmentation

BUSINESS IMPACT

Automated QA pipeline reduces human error and accelerates quality assurance for global product lines • Ensures regulatory compliance for CE marks, age grades, safety warnings • Scalable solution for diverse product portfolios

Floor Plan Agentic Platform



Technology: Next.js, Pytorch, Huggingface, vLLM, Autogen, Amazon Web Service

Challenges

- Manual floor plan design takes hours or days for architects
- Manual inspection prone to human error and oversight

Solutions

- Multi-agent system for floor plan generation
- LLM based layouts compliance against requirements with detailed feedback
- Direct output to CAD format for ease integration

Key Results

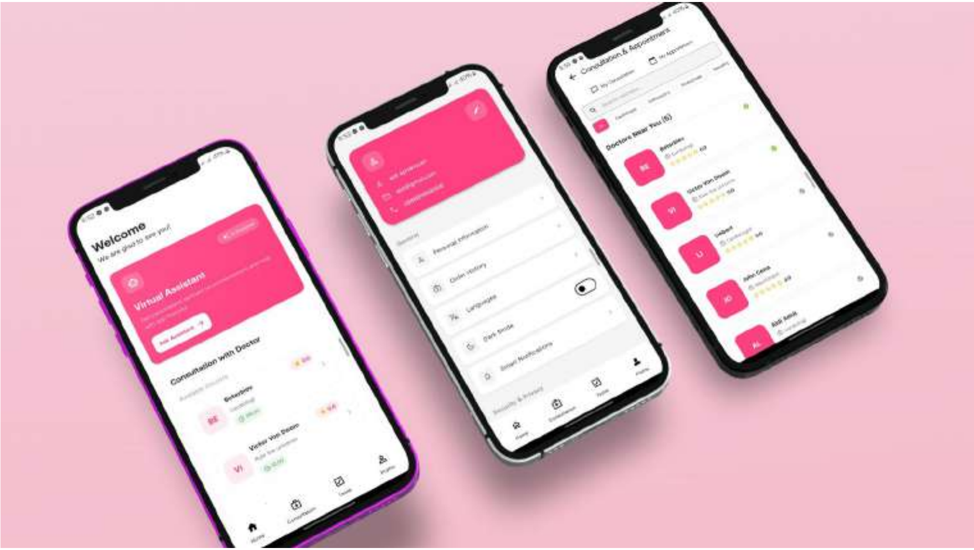
- Floor Plan Generation IoU: **88.24%**
- Passed all given scenarios during end-to-end test for floor plan generation agents
- Compliance Accuracy on test data: **94.65%**

BUSINESS IMPACT

Reduced design cycle time from days to minutes, enabling to scale operations and take on more construction projects simultaneously • Automated compliance validation minimizes costly rework and accelerates time-to-market for boarding house and co-living developments.

Telehealth

Mobile Healthcare Consultation and Telemetry App



Technology: Flutter, LLM, Langchain, Flask, Google Cloud Platform

Github: https://github.com/adhitaazizi/TechTalent-Academy

Challenges

- Patients in rural or underserved areas lack proximity to specialized healthcare providers, leading to delayed diagnoses and treatments
- Patients with chronic conditions require consistent monitoring and follow-ups, which becomes difficult when in-person visits are mandatory for every interaction

Solutions

Cross-platform mobile application with secure video consultations, AI virtual assistant, automated scheduling, and medical record management to connect patients with healthcare providers remotely.

Key Results

Eliminated geographic barriers for underserved populations, reduced appointment wait times through streamlined scheduling, and enabled continuous remote monitoring for chronic condition management.

Tech Talent Academy

AI-Enhanced Learning Management System

Technology: React Native, Expo, NativeWind, AI APIs, Google Speech-to-Text, Google Cloud Platform

Github: https://github.com/adhitaazizi/TechTalent-Academy

Challenges

Traditional learning platforms fail to provide personalized guidance, practical interview preparation, and adaptive content. Users struggle with generic advice that doesn't address their specific needs.

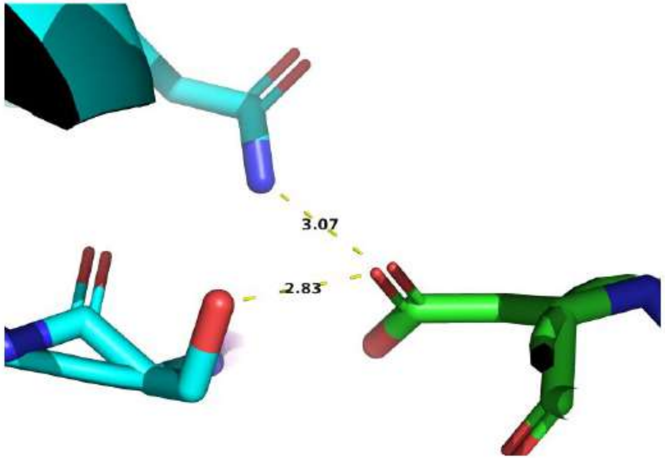
Solutions

AI-powered application with personalized guidance, mock interview simulation with voice recording and real-time transcription, performance analysis with improvement suggestions, and personalized hobby recommendations.

Key Results

Cross-platform mobile app built with React Native and Expo, integrated AI APIs for natural language processing, implemented speech-to-text for interview simulation, and created intuitive UI with modern animations.

Antibody-Antigen Affinity Prediction



Technology: Pytorch, PyG, Graph Neural Network, Protein 3D Structure Prediction, Protein Language Model, Biopython

Github: https://github.com/adhitaazizi/chewy

Challenges

Traditional wet-lab experimentation for determining antibody-antigen binding affinity is time-consuming, expensive, and limits the speed of therapeutic antibody development and drug discovery.

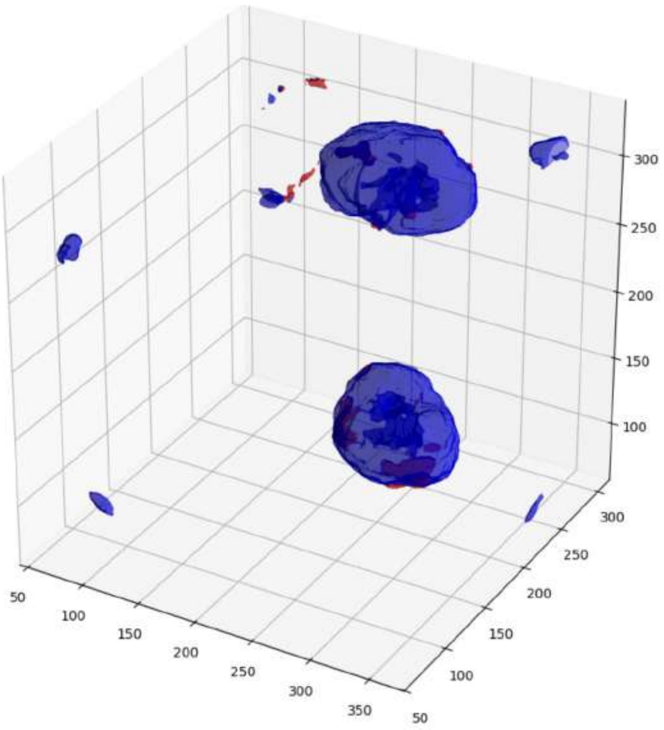
Solutions

Deep learning models using Protein Language Model (PLM), Protein 3D structure prediction and graph neural networks to analyze protein structures and predict binding affinity computationally, enabling rapid screening of antibody candidates.

Key Results

- Unrelated Complex evaluation: **0.64**
- Local Perturbation evaluation: **0.66**

3D Kidney Segmentation



Challenges

Manual segmentation of kidney structures from CT/MRI scans is time-intensive for radiologists, prone to human error, and limits the ability to provide quantitative analysis for diagnostic decision-making.

Solutions

3D convolutional neural networks using U-Net architecture variants with medical imaging libraries to automatically segment kidney structures from volumetric medical scans.

Key Results

- Dice Score: **0.86**
- IoU: **0.84**

Technology: Pytorch, 3D Segmentation, CT Images, FC-CNN

Github: https://github.com/adhitaazizi/computer-vision-project

Let’s Collaborate

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